Lesson notes · for you

Keep connected things safe

IoT · Level 4 · Lesson 3

Materials
A connected device your family owns, and their permission; Its settings app or page, or its manual; Paper and a pen for the checklist

Learning goals

- Explain how attackers take over devices with default passwords, and what a botnet is.
- Secure a device: a new strong password, updates, and only the features you use.
- Decide what a device should and shouldn't collect.

Activity: Device safety check

1. Choose one connected device at home, with its owner's permission: the Wi-Fi router, a smart TV, a camera or a smart speaker. Find its settings app or page, or its manual.
2. Ask the owner whether the password is still the one on the box or the label. If it is, help them choose a long passphrase. Don't write the password on your sheet.
3. In the settings, look for “automatic updates” or “software update”. Note whether it is on, and when the last update was.
4. Look for remote access, the microphone, location and anything else that is switched on. Ask: does anyone use this? Suggest switching off what nobody needs.
5. List what the device collects (pictures, sound, location, what you watch) and mark what its job doesn't need. Share one change with your family.

Only check devices you or your family own, with the owner's permission. Never put a password in your notes or photos, and never try to log in to anyone else's device.

Discussion

- Why did Mirai find so many devices so easily?
- Which feature at home would you switch off, and why?
- Who should keep devices safe: the maker, the shop or the owner?

Check yourself

- Name two habits that would have stopped Mirai.

Try it this week
Device check: with the owner's permission, pick one device at home, like the Wi-Fi router, and check that its password was changed, updates are on, and nothing is switched on that nobody uses.

Worksheet

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Name _____ Class _____ Date _____

1 **The school's new entrance camera can also record sound, but nobody needs it. What should the school do?**

◯ Switch the sound off ◯ Leave the sound on, just in case

◯ Record sound, but let only staff listen to it

2 **Match each habit to the attack it stops.**

Change the default password	• • • •	Attacks on known holes in old software
Turn on automatic updates	• • • •	Strangers reaching the device from the internet
Switch off remote access	• • • •	Private conversations leaking
Put devices on a guest network	• • • •	Programs trying factory passwords
Switch off the microphone	• • • •	A hijacked camera reaching the family laptop

3 **Adding symbols, as in P@ssw0rd!, turns a common password into a strong one.**

◯ True ◯ False

4 **Grace's family is setting up a new smart TV. Which steps make it safer?**

☐ Leave the microphone on, in case they want voice search one day

☐ Keep the factory password, so the shop can help if it breaks

☐ Set a new, long password for its account ☐ Turn on automatic updates

☐ Put it on the guest Wi-Fi

5 **Your neighbour's Wi-Fi camera probably still uses the password admin. What should you do?**

◯ Log in to check, then tell them ◯ Post a warning in the street's WhatsApp group

◯ Tell them privately how to change it

✎ **Which connected device at home would you check first, and why? What does it collect that its job may not need?**

Check your answers

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1 **Switch the sound off**

Data that's never recorded can never leak, and a feature that's switched off can't be misused.

2 **Change the default password — Programs trying factory passwords**
Turn on automatic updates — Attacks on known holes in old software
Switch off remote access — Strangers reaching the device from the internet
Put devices on a guest network — A hijacked camera reaching the family laptop
Switch off the microphone — Private conversations leaking

Each habit closes one door. Together they stop most attacks on devices at home and at school, and none of them costs money.

3 **False**

Guessing programs try common words with swaps like @ for a and 0 for o almost at once.

4 **Set a new, long password for its account; Turn on automatic updates; Put it on the guest Wi-Fi**

A new password, updates and a separate network close the main doors.

5 **Tell them privately how to change it**

Logging in to someone else's device, even to help, is against the law in many countries and breaks their trust.